Practice 4 — Functions (Mappings)

Solutions to Selected Problems

Problems 2, 4, 5, 6, 10, 12, 19, 22, 24, 25

Discrete Mathematics — QYSU

Contents

1 Problem 2 — Domain and Range of Real Functions 2

2 Problem 4 — Composition of Functions 5

3 Problem 5 — Inverse Functions 7

4 Problem 6 — Injectivity, Surjectivity, Bijectivity 9

5 Problem 10 — Kernel of a Mapping is an Equivalence Relation 11

6 Problem 12 — $f(x) = \frac{x}{2x+1}$ on $\mathbb{Q}^+$: Injective but Not Surjective 12

7 Problem 19 — $f(A \cap B) = f(A) \cap f(B)$ Iff f Is Injective 14

8 Problem 22 — Disjoint Subsets with Equal Images 16

9 Problem 24 — Preimages of Disjoint Sets 17

10 Problem 25 — $f(f^{-1}(Y)) \subseteq Y$ 18

1 Problem 2 — Domain and Range of Real Functions

Theory Recap: Domain and Range

For a function $f : \mathbb{R} \to \mathbb{R}$ given by a formula $f(x) = \ldots$:

- The **domain** $\mathrm{Dom}(f)$ is the set of all $x \in \mathbb{R}$ for which the formula is defined (no division by zero, no square root of a negative number, etc.).
- The **range** (image) $\mathrm{Ran}(f) = \{f(x) \mid x \in \mathrm{Dom}(f)\}$ is the set of all values actually attained by f.

Common restrictions:

Expression	Restriction	Reason
$\sqrt{g(x)}$	$g(x) \geq 0$	square root of negative undefined in $\mathbb{R}$
$1/g(x)$	$g(x) \neq 0$	division by zero undefined
$1/\sqrt{g(x)}$	$g(x) > 0$	both restrictions combined

Problem Statement

Let $f : \mathbb{R} \to \mathbb{R}$. Find the domain and range of:

(a) $f(x) = x^2 + 3$

(b) $f(x) = \sqrt{x+2}$

(c) $f(x) = \dfrac{1}{\sqrt{x-2}}$

(d) $f(x) = |x|$

(e) $f(x) = \dfrac{1}{x^2+2}$

(f) $f(x) = \dfrac{1}{x^2-2}$

Solution

Two-step method for each mapping:

1. **Domain check:** determine where the formula is defined.
2. **Range check:** determine which outputs are actually achieved.

(a) $f(x) = x^2 + 3$**.**

No restrictions on x: every real number can be squared.

$$\mathrm{Dom}(f) = \mathbb{R}.$$

Since $x^2 \geq 0$ for all $x \in \mathbb{R}$, we have $f(x) = x^2 + 3 \geq 3$. The minimum value 3 is attained at $x = 0$. As $x \to \pm\infty$, $f(x) \to +\infty$, and by continuity every value ≥ 3 is attained.

$$\mathrm{Ran}(f) = [3, +\infty).$$

$\mathrm{Dom}(f) = \mathbb{R}, \quad \mathrm{Ran}(f) = [3, +\infty).$

(b) $f(x) = \sqrt{x+2}$**.**

We need $x + 2 \geq 0$, i.e. $x \geq -2$.

$$\mathrm{Dom}(f) = [-2, +\infty).$$

At $x = -2$: $f(-2) = 0$. As $x \to +\infty$, $f(x) \to +\infty$. Since $\sqrt{\cdot}$ is continuous and non-negative:

$$\mathrm{Ran}(f) = [0, +\infty).$$

$\mathrm{Dom}(f) = [-2, +\infty), \quad \mathrm{Ran}(f) = [0, +\infty).$

(c) $f(x) = \dfrac{1}{\sqrt{x-2}}$**.**

We need $x - 2 > 0$ (strict inequality: the square root must be positive since it appears in the denominator), i.e. $x > 2$.

$$\mathrm{Dom}(f) = (2, +\infty).$$

As $x \to 2^+$: $\sqrt{x-2} \to 0^+$, so $f(x) \to +\infty$. As $x \to +\infty$: $\sqrt{x-2} \to +\infty$, so $f(x) \to 0^+$. Since f is continuous and strictly decreasing on $(2, +\infty)$, it takes every positive value:

$$\mathrm{Ran}(f) = (0, +\infty).$$

$\mathrm{Dom}(f) = (2, +\infty), \quad \mathrm{Ran}(f) = (0, +\infty).$

(d) $f(x) = |x|$**.**

The absolute value is defined for all real numbers.

$$\mathrm{Dom}(f) = \mathbb{R}.$$

Since $|x| \geq 0$ for all x, and $|0| = 0$, and $|x| \to +\infty$ as $x \to \pm\infty$:

$$\mathrm{Ran}(f) = [0, +\infty).$$

$\mathrm{Dom}(f) = \mathbb{R}, \quad \mathrm{Ran}(f) = [0, +\infty).$

(e) $f(x) = \dfrac{1}{x^2+2}$**.**

Since $x^2 + 2 \geq 2 > 0$ for all $x \in \mathbb{R}$, there is no division by zero.

$$\mathrm{Dom}(f) = \mathbb{R}.$$

Since $x^2 + 2 \geq 2$, we have $f(x) = \frac{1}{x^2+2} \leq \frac{1}{2}$. The maximum $\frac{1}{2}$ is attained at $x = 0$. As $x \to \pm\infty$, $f(x) \to 0^+$, but $f(x) > 0$ for all x. By continuity, every value in $(0, \frac{1}{2}]$ is attained.

$$\mathrm{Ran}(f) = \left(0, \tfrac{1}{2}\right].$$

$\mathrm{Dom}(f) = \mathbb{R}, \quad \mathrm{Ran}(f) = \left(0,\ \frac{1}{2}\right].$

(f) $f(x) = \dfrac{1}{x^2 - 2}$.

We need $x^2 - 2 \neq 0$, i.e. $x \neq \pm\sqrt{2}$.

$$\mathrm{Dom}(f) = \mathbb{R} \setminus \{-\sqrt{2},\ \sqrt{2}\}.$$

To find the range, set $y = \frac{1}{x^2-2}$ and solve for x^2:

$$y(x^2 - 2) = 1 \implies x^2 = \frac{1}{y} + 2 = \frac{1 + 2y}{y}.$$

We need $y \neq 0$ and $x^2 = \frac{1+2y}{y} \geq 0$.

Analyzing the sign of $\frac{1+2y}{y}$:

- $y > 0$: need $1 + 2y > 0$, which holds for all $y > 0$. So $y \in (0, +\infty)$.
- $y < 0$: need $1 + 2y < 0$ (numerator and denominator both negative), i.e. $y < -\frac{1}{2}$. So $y \in (-\infty, -\frac{1}{2})$.

Sign analysis. Solving $x^2 = \frac{1+2y}{y} \geq 0$ requires the numerator and denominator to have the same sign. For $y > 0$: $1 + 2y > 0$ automatically, so all $y > 0$ work. For $y < 0$: we need $1+2y < 0$, i.e. $y < -\frac{1}{2}$. Hence $y \in (-\infty, -\frac{1}{2}) \cup (0, +\infty)$, plus we must check the boundary.

At $x = 0$: $f(0) = \frac{1}{-2} = -\frac{1}{2}$, so $y = -\frac{1}{2}$ is attained.

$$\mathrm{Ran}(f) = \left(-\infty,\ -\tfrac{1}{2}\right] \cup (0, +\infty).$$

$\mathrm{Dom}(f) = \mathbb{R} \setminus \{-\sqrt{2},\ \sqrt{2}\}, \quad \mathrm{Ran}(f) = \left(-\infty,\ -\frac{1}{2}\right] \cup (0, +\infty).$

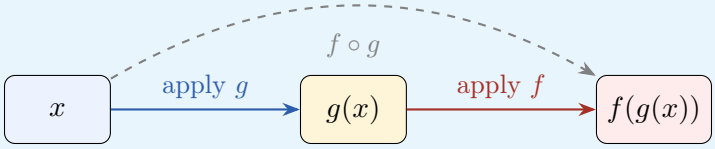
2 Problem 4 — Composition of Functions

Theory Recap: Function Composition

Given $f, g : \mathbb{R} \to \mathbb{R}$, the **composition** $f \circ g$ is defined by:

$$(f \circ g)(x) = f\big(g(x)\big).$$

Read right to left: first apply g, then apply f to the result.



Warning: In general, $f \circ g \neq g \circ f$ — composition is not commutative!

Problem Statement

Let $f, g : \mathbb{R} \to \mathbb{R}$. Find $f \circ g$ and $g \circ f$ if:

(a) $f(x) = x^2 + 1, \quad g(x) = x + 3$

(b) $f(x) = \sqrt{x^2 + 2}, \quad g(x) = x^2 + 3$

(c) $f(x) = \dfrac{1}{x}, \quad g(x) = 2x + 3$

Solution

(a) $f(x) = x^2 + 1$, $g(x) = x + 3$.

$$(f \circ g)(x) = f(g(x)) = f(x+3) = (x+3)^2 + 1 = x^2 + 6x + 9 + 1 = x^2 + 6x + 10.$$

$$(g \circ f)(x) = g(f(x)) = g(x^2+1) = (x^2+1) + 3 = x^2 + 4.$$

$$\boxed{(f \circ g)(x) = x^2 + 6x + 10, \qquad (g \circ f)(x) = x^2 + 4.}$$

(b) $f(x) = \sqrt{x^2 + 2}$, $g(x) = x^2 + 3$.

$$(f \circ g)(x) = f(g(x)) = f(x^2+3) = \sqrt{(x^2+3)^2 + 2} = \sqrt{x^4 + 6x^2 + 9 + 2} = \sqrt{x^4 + 6x^2 + 11}.$$

$$(g \circ f)(x) = g(f(x)) = g\left(\sqrt{x^2+2}\right) = \left(\sqrt{x^2+2}\right)^2 + 3 = x^2 + 2 + 3 = x^2 + 5.$$

$$\boxed{(f \circ g)(x) = \sqrt{x^4 + 6x^2 + 11}, \qquad (g \circ f)(x) = x^2 + 5.}$$

(c) $f(x) = \dfrac{1}{x}$, $g(x) = 2x + 3$.

$$(f \circ g)(x) = f(g(x)) = f(2x+3) = \frac{1}{2x+3}, \qquad x \neq -\tfrac{3}{2}.$$

$$(g \circ f)(x) = g(f(x)) = g\left(\frac{1}{x}\right) = \frac{2}{x} + 3 = \frac{2+3x}{x}, \qquad x \neq 0.$$

Domain check. After computing the formula, verify where it is defined—the domain of a composition $(g \circ f)(x)$ requires $x \in \mathrm{dom}(f)$ and $f(x) \in \mathrm{dom}(g)$.

$$(f \circ g)(x) = \frac{1}{2x+3}\ (x \neq -\tfrac{3}{2}), \qquad (g \circ f)(x) = \frac{2+3x}{x}\ (x \neq 0).$$

3 Problem 5 — Inverse Functions

Theory Recap: Inverse Function

If $f : A \to B$ is a bijection, then f has an **inverse function** $f^{-1} : B \to A$ satisfying:

$$f^{-1}(y) = x \iff f(x) = y.$$

Algorithm to find f^{-1}:

1. Write $y = f(x)$.
2. Solve for x in terms of y.
3. Swap the names: the expression for x in terms of y is $f^{-1}(y)$.

Verification: $f^{-1}(f(x)) = x$ and $f(f^{-1}(y)) = y$.

Problem Statement

Find the inverse relation for:

(a) $f(x) = \dfrac{x+4}{2}$

(b) $f(x) = x^3$

(c) $f(x) = \dfrac{x-2}{x+3}$

Solution

(a) $f(x) = \dfrac{x+4}{2}$.

Set $y = \frac{x+4}{2}$. Solve for x:

$$2y = x + 4 \implies x = 2y - 4.$$

$$\boxed{f^{-1}(y) = 2y - 4.}$$

Verification: $f(f^{-1}(y)) = f(2y-4) = \frac{(2y-4)+4}{2} = \frac{2y}{2} = y$. ✓

$f^{-1}(f(x)) = f^{-1}\left(\frac{x+4}{2}\right) = 2 \cdot \frac{x+4}{2} - 4 = x + 4 - 4 = x$. ✓

$f^{-1}(x) = 2x - 4.$

(b) $f(x) = x^3$.

Set $y = x^3$. Solve for x:

$$x = \sqrt[3]{y} = y^{1/3}.$$

$$\boxed{f^{-1}(y) = \sqrt[3]{y}.}$$

Verification: $f(f^{-1}(y)) = (\sqrt[3]{y})^3 = y$. ✓ $\quad f^{-1}(f(x)) = \sqrt[3]{x^3} = x$. ✓

$f^{-1}(x) = \sqrt[3]{x}.$

(c) $f(x) = \dfrac{x-2}{x+3}, \qquad x \neq -3.$

Set $y = \frac{x-2}{x+3}$. Solve for x:

$$\begin{aligned} y(x+3) &= x-2 \\ yx + 3y &= x-2 \\ yx - x &= -2-3y \\ x(y-1) &= -(2+3y) \\ x &= \frac{-(2+3y)}{y-1} = \frac{2+3y}{1-y}, \qquad y \neq 1. \end{aligned}$$

Note that $f(x) = 1$ has no solution (it would require $x-2 = x+3$, i.e. $-2 = 3$), confirming $1 \notin \mathrm{Ran}(f)$.

$$\boxed{f^{-1}(y) = \frac{2+3y}{1-y}, \qquad y \neq 1.}$$

Verification: $f(f^{-1}(y)) = f\left(\frac{2+3y}{1-y}\right) = \frac{\frac{2+3y}{1-y}-2}{\frac{2+3y}{1-y}+3} = \frac{2+3y-2(1-y)}{2+3y+3(1-y)} = \frac{2+3y-2+2y}{2+3y+3-3y} = \frac{5y}{5} = y.$ ✓

$$f^{-1}(x) = \frac{2+3x}{1-x}, \quad x \neq 1.$$

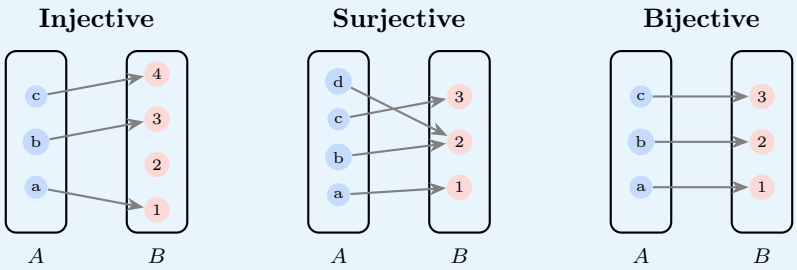
Note. Here the inverse relation happens to be a function because each mapping is bijective onto its range. In general, the inverse *relation* always exists, but it is a *function* only when the original mapping is injective.

4 Problem 6 — Injectivity, Surjectivity, Bijectivity

Theory Recap: Injective, Surjective, Bijective

Let $f : A \to B$.

- **Injective** (one-to-one): $f(x_1) = f(x_2) \implies x_1 = x_2$. Equivalently, $x_1 \neq x_2 \implies f(x_1) \neq f(x_2)$. *Horizontal line test:* every horizontal line meets the graph at most once.
- **Surjective** (onto): $\forall\, y \in B,\ \exists\, x \in A : f(x) = y$. Equivalently, $\mathrm{Ran}(f) = B$.
- **Bijective**: both injective and surjective.



Problem Statement

Determine which of the following mappings $f : \mathbb{R} \to \mathbb{R}$ are injective, surjective, or bijective:

(a) $f(x) = |x|$
(b) $f(x) = x^2 + 4$
(c) $f(x) = x^3 + 6$
(d) $f(x) = x + |x|$
(e) $f(x) = x(x-2)(x+2)$

Solution

Efficient tests for injectivity and surjectivity:

- $|x|$: symmetry ($f(x) = f(-x)$) immediately shows non-injectivity.
- $x^2 + c$: parabola has a minimum, so not surjective onto $\mathbb{R}$.
- $x^3 + c$: strictly monotone, hence injective; end behavior gives surjectivity.
- $x + |x|$: rewrite piecewise first, then check each piece.
- $x^3 - 4x$: look for repeated outputs (local max/min) and use end behavior for surjectivity.

(a) $f(x) = |x|$**.**

Not injective: $f(1) = |1| = 1 = |-1| = f(-1)$, but $1 \neq -1$.

Not surjective: $|x| \geq 0$ for all x, so no x satisfies $f(x) = -1$. The range is $[0, +\infty) \neq \mathbb{R}$.

$f(x) = |x|$: neither injective nor surjective (hence not bijective).

(b) $f(x) = x^2 + 4$**.**

Not injective: $f(1) = 5 = f(-1)$, but $1 \neq -1$.

Not surjective: $x^2 + 4 \geq 4$ for all x, so no x satisfies $f(x) = 0$. The range is $[4, +\infty) \neq \mathbb{R}$.

$f(x) = x^2 + 4$: neither injective nor surjective (hence not bijective).

(c) $f(x) = x^3 + 6$**.**

Injective: Suppose $f(x_1) = f(x_2)$. Then $x_1^3 + 6 = x_2^3 + 6$, so $x_1^3 = x_2^3$. Since the cube function is strictly increasing on $\mathbb{R}$, this implies $x_1 = x_2$.

Surjective: Given any $y \in \mathbb{R}$, we need $x^3 + 6 = y$, i.e. $x = \sqrt[3]{y - 6}$. Since the cube root is defined for all real numbers, such x always exists.

$f(x) = x^3 + 6$: **bijective** (both injective and surjective).

(d) $f(x) = x + |x|$**.**

First, let us simplify:

$$f(x) = x + |x| = \begin{cases} 2x & \text{if } x \geq 0, \\ 0 & \text{if } x < 0. \end{cases}$$

Not injective: $f(-1) = 0 = f(-2)$, but $-1 \neq -2$. (In fact, $f(x) = 0$ for all $x < 0$.)

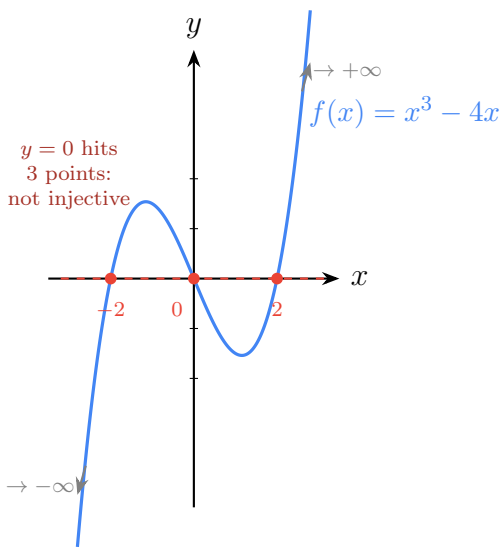
Not surjective: For $x \geq 0$, $f(x) = 2x \geq 0$. For $x < 0$, $f(x) = 0$. So $f(x) \geq 0$ for all x, and no x satisfies $f(x) = -1$. The range is $[0, +\infty) \neq \mathbb{R}$.

$f(x) = x + |x|$: neither injective nor surjective (hence not bijective).

(e) $f(x) = x(x-2)(x+2) = x^3 - 4x$**.**

Not injective: $f(0) = 0$ and $f(2) = 2 \cdot 0 \cdot 4 = 0$, so $f(0) = f(2) = 0$ but $0 \neq 2$.

Surjective: $f(x) = x^3 - 4x$ is a cubic polynomial with positive leading coefficient. As $x \to -\infty$, $f(x) \to -\infty$, and as $x \to +\infty$, $f(x) \to +\infty$. By the Intermediate Value Theorem, every $y \in \mathbb{R}$ is attained.



$f(x) = x(x-2)(x+2)$: surjective but **not injective** (hence not bijective).

5 Problem 10 — Kernel of a Mapping is an Equivalence Relation

Theory Recap: Kernel of a Mapping

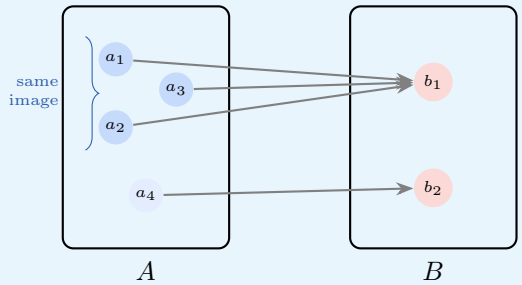
Let $f : A \to B$ be a mapping. The **kernel** of f is the relation on A defined by:

$$\mathrm{Ker}(f) = \{(a_1, a_2) \in A \times A \mid f(a_1) = f(a_2)\}.$$

We write $a_1 \sim a_2$ to mean $(a_1, a_2) \in \mathrm{Ker}(f)$, i.e. $f(a_1) = f(a_2)$.

An **equivalence relation** on A is a relation $\sim$ that is:

1. **Reflexive:** $a \sim a$ for all $a \in A$.
2. **Symmetric:** $a \sim b \implies b \sim a$.
3. **Transitive:** $a \sim b \wedge b \sim c \implies a \sim c$.



The equivalence classes of $\mathrm{Ker}(f)$ are exactly the **fibers** $f^{-1}(\{b\}) = \{a \in A \mid f(a) = b\}$ for $b \in \mathrm{Ran}(f)$.

Problem Statement

Prove that for any mapping $f : A \to B$, its kernel $\mathrm{Ker}(f)$ is an equivalence relation on A.

Solution

We must verify the three properties of an equivalence relation. Recall: $a_1 \sim a_2 \iff f(a_1) = f(a_2)$.

1. Reflexivity. Let $a \in A$. Then $f(a) = f(a)$ (equality is reflexive), so $a \sim a$. ✓

2. Symmetry. Suppose $a_1 \sim a_2$, i.e. $f(a_1) = f(a_2)$. Then $f(a_2) = f(a_1)$ (equality is symmetric), so $a_2 \sim a_1$. ✓

3. Transitivity. Suppose $a_1 \sim a_2$ and $a_2 \sim a_3$, i.e. $f(a_1) = f(a_2)$ and $f(a_2) = f(a_3)$. Then $f(a_1) = f(a_3)$ (equality is transitive), so $a_1 \sim a_3$. ✓

Since $\mathrm{Ker}(f)$ is reflexive, symmetric, and transitive, it is an equivalence relation on A. □

The kernel $\mathrm{Ker}(f)$ inherits reflexivity, symmetry, and transitivity from the equality relation "=" on B, so it is an equivalence relation on A.

Concrete example. If $f\colon \{1,2,3,4\} \to \{a,b\}$ with $f(1) = f(3) = a$ and $f(2) = f(4) = b$, then:

- The equivalence class of 1 is $f^{-1}(\{a\}) = \{1,3\}$.
- The equivalence class of 2 is $f^{-1}(\{b\}) = \{2,4\}$.
- The partition is $\big\{\{1,3\},\ \{2,4\}\big\}$.

6 Problem 12 — $f(x) = \frac{x}{2x+1}$ on $\mathbb{Q}^+$: Injective but Not Surjective

Theory Recap: Injectivity and Surjectivity — Proof Techniques

To prove injectivity: Assume $f(x_1) = f(x_2)$ and show $x_1 = x_2$ (direct proof), or assume $x_1 \neq x_2$ and show $f(x_1) \neq f(x_2)$ (contrapositive).
To disprove surjectivity: Find a specific y in the codomain for which $f(x) = y$ has no solution in the domain.
To prove surjectivity: Given arbitrary y in the codomain, solve $f(x) = y$ and verify x lies in the domain.

Problem Statement

Let $f : \mathbb{Q}^+ \to \mathbb{Q}^+$ be defined by $f(x) = \dfrac{x}{2x+1}$. Prove that f is injective but not surjective.

Solution

Intuition. Rewriting $f(x) = \dfrac{1}{2 + 1/x}$ shows that as x increases on $\mathbb{Q}^+$, the denominator $2 + 1/x$ decreases, so $f(x)$ increases. A strictly increasing function is always injective.

Injectivity. Let $x_1, x_2 \in \mathbb{Q}^+$ and suppose $f(x_1) = f(x_2)$:

$$\frac{x_1}{2x_1 + 1} = \frac{x_2}{2x_2 + 1}.$$

Cross-multiplying (the denominators are positive since $x_1, x_2 > 0$):

$$x_1(2x_2 + 1) = x_2(2x_1 + 1).$$

Expanding:

$$2x_1x_2 + x_1 = 2x_1x_2 + x_2.$$

Subtracting $2x_1x_2$ from both sides:

$$x_1 = x_2.$$

Therefore f is injective. $\square$

Not surjective. We claim that $y = \frac{1}{2}$ is in $\mathbb{Q}^+$ but not in $\mathrm{Ran}(f)$.
Suppose for contradiction that there exists $x \in \mathbb{Q}^+$ with $f(x) = \frac{1}{2}$:

$$\frac{x}{2x+1} = \frac{1}{2} \implies 2x = 2x + 1 \implies 0 = 1.$$

This is a contradiction. Therefore no $x \in \mathbb{Q}^+$ satisfies $f(x) = \frac{1}{2}$.

More generally, for any $x > 0$:

$$f(x) = \frac{x}{2x+1} = \frac{1}{2 + \frac{1}{x}} < \frac{1}{2},$$

since $\frac{1}{x} > 0$, so $2 + \frac{1}{x} > 2$, and thus $\frac{1}{2+\frac{1}{x}} < \frac{1}{2}$. Therefore $\mathrm{Ran}(f) \subseteq \left(0, \frac{1}{2}\right)$, so f is not surjective onto $\mathbb{Q}^+$. $\square$

$f(x) = \frac{x}{2x+1}$ on $\mathbb{Q}^+$ is **injective** (by cross-multiplication) but **not surjective** (e.g., $y = \frac{1}{2}$ is not in the range, since $f(x) < \frac{1}{2}$ for all $x > 0$).

7 Problem 19 — $f(A \cap B) = f(A) \cap f(B)$ Iff f Is Injective

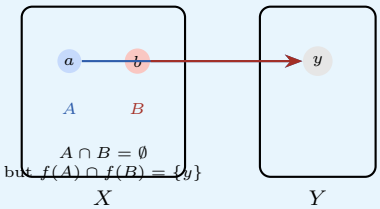
Theory Recap: Images of Intersections

For any mapping $f : X \to Y$ and subsets $A, B \subseteq X$:

$$f(A \cap B) \subseteq f(A) \cap f(B) \qquad \text{always holds.}$$

However, equality $f(A \cap B) = f(A) \cap f(B)$ does **not** hold in general. It holds for *all* $A, B \subseteq X$ if and only if f is **injective**.

Non-injective f



Problem Statement

For which mappings does $f(A \cap B) = f(A) \cap f(B)$ hold for all subsets A, B of the domain?
Answer: injective mappings. Prove both directions.

Solution

Why this matters. If f is not injective—say $f(1) = f(2) = y$—then $f(\{1\}) \cap f(\{2\}) = \{y\} \neq \emptyset$, but $\{1\} \cap \{2\} = \emptyset$. So $f(A \cap B) \neq f(A) \cap f(B)$ in general. Injectivity is exactly the condition that prevents this.

Direction 1: If f is injective, then $f(A \cap B) = f(A) \cap f(B)$.

The inclusion $f(A \cap B) \subseteq f(A) \cap f(B)$ holds for any function (if $x \in A \cap B$, then $f(x) \in f(A)$ and $f(x) \in f(B)$).

We prove the reverse inclusion $f(A) \cap f(B) \subseteq f(A \cap B)$.

Let $y \in f(A) \cap f(B)$. Then:

- $y \in f(A)$: there exists $a \in A$ with $f(a) = y$.
- $y \in f(B)$: there exists $b \in B$ with $f(b) = y$.

Since $f(a) = y = f(b)$ and f is injective, we conclude $a = b$. Call this common element $c = a = b$. Then $c \in A$ and $c \in B$, so $c \in A \cap B$. Therefore $y = f(c) \in f(A \cap B)$.

This shows $f(A) \cap f(B) \subseteq f(A \cap B)$, completing the proof of equality. $\square$

Direction 2: If $f(A \cap B) = f(A) \cap f(B)$ for all A, B, then f is injective.

We prove the contrapositive: if f is not injective, then there exist A, B with $f(A \cap B) \neq f(A) \cap f(B)$.

Suppose f is not injective. Then there exist $x_1 \neq x_2$ in the domain with $f(x_1) = f(x_2)$. Set $A = \{x_1\}$ and $B = \{x_2\}$. Then:

- $A \cap B = \emptyset$ (since $x_1 \neq x_2$), so $f(A \cap B) = f(\emptyset) = \emptyset$.
- $f(A) = \{f(x_1)\}$ and $f(B) = \{f(x_2)\} = \{f(x_1)\}$, so $f(A) \cap f(B) = \{f(x_1)\} \neq \emptyset$.

Therefore $f(A \cap B) \neq f(A) \cap f(B)$. $\square$

$f(A \cap B) = f(A) \cap f(B)$ holds for all A, B if and only if f is injective.

8 Problem 22 — Disjoint Subsets with Equal Images

Theory Recap: Images of Disjoint Sets

If $f : A \to B$ is **not** injective, then different elements can map to the same image. This means disjoint subsets of the domain can have overlapping — or even equal — images.

Problem Statement

Is it possible for $f : A \to B$ and disjoint non-empty $X_1, X_2 \subseteq A$ (i.e. $X_1 \cap X_2 = \emptyset$, $X_1 \neq \emptyset$, $X_2 \neq \emptyset$) to have $f(X_1) = f(X_2)$?

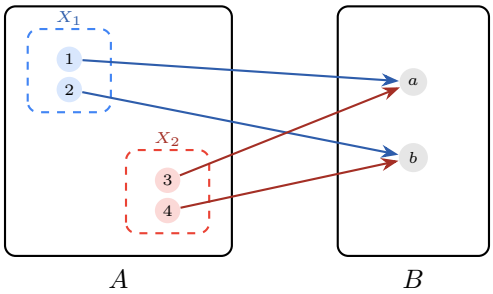
Solution

Yes, this is possible. We give a concrete example.

Let $A = \{1, 2, 3, 4\}$, $B = \{a, b\}$, and define f by:

$$f(1) = a, \quad f(2) = b, \quad f(3) = a, \quad f(4) = b.$$

Take $X_1 = \{1, 2\}$ and $X_2 = \{3, 4\}$.



Then:

- $X_1 \cap X_2 = \{1, 2\} \cap \{3, 4\} = \emptyset$. ✓
- $f(X_1) = \{f(1), f(2)\} = \{a, b\}$.
- $f(X_2) = \{f(3), f(4)\} = \{a, b\}$.
- $f(X_1) = f(X_2) = \{a, b\}$. ✓

Yes. Disjoint non-empty subsets can have equal images when f is not injective. Example: $f(1) = a,\ f(2) = b,\ f(3) = a,\ f(4) = b$ with $X_1 = \{1, 2\}$, $X_2 = \{3, 4\}$.

9 Problem 24 — Preimages of Disjoint Sets

Theory Recap: Preimages Preserve Set Operations

For any function $f : A \to B$ and subsets $Y_1, Y_2 \subseteq B$, the **preimage** $f^{-1}(Y) = \{x \in A \mid f(x) \in Y\}$ satisfies:

- $f^{-1}(Y_1 \cup Y_2) = f^{-1}(Y_1) \cup f^{-1}(Y_2)$
- $f^{-1}(Y_1 \cap Y_2) = f^{-1}(Y_1) \cap f^{-1}(Y_2)$
- $f^{-1}(Y_1 \setminus Y_2) = f^{-1}(Y_1) \setminus f^{-1}(Y_2)$

These hold for *any* function — no injectivity or surjectivity is required!

Problem Statement

Is it true that for $f : A \to B$, if $Y_1 \cap Y_2 = \emptyset$, then $f^{-1}(Y_1) \cap f^{-1}(Y_2) = \emptyset$?

Solution

Yes, this is always true.

Proof. Suppose $Y_1 \cap Y_2 = \emptyset$. We use the identity $f^{-1}(Y_1 \cap Y_2) = f^{-1}(Y_1) \cap f^{-1}(Y_2)$. Since $Y_1 \cap Y_2 = \emptyset$:

$$f^{-1}(Y_1) \cap f^{-1}(Y_2) = f^{-1}(Y_1 \cap Y_2) = f^{-1}(\emptyset) = \emptyset.$$

Alternatively, we can prove it directly. Suppose for contradiction that there exists $x \in f^{-1}(Y_1) \cap f^{-1}(Y_2)$. Then:

- $x \in f^{-1}(Y_1) \implies f(x) \in Y_1$.
- $x \in f^{-1}(Y_2) \implies f(x) \in Y_2$.

So $f(x) \in Y_1 \cap Y_2 = \emptyset$, which is a contradiction since no element belongs to the empty set. $\square$

Yes. If $Y_1 \cap Y_2 = \emptyset$, then $f^{-1}(Y_1) \cap f^{-1}(Y_2) = \emptyset$. This follows from the fact that preimages preserve intersections.

Contrast with images. Preimages *always* preserve intersections: $f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$. Images do *not*, unless f is injective (see Problem 19). This asymmetry is a fundamental feature of functions.

10 Problem 25 — $f(f^{-1}(Y)) \subseteq Y$

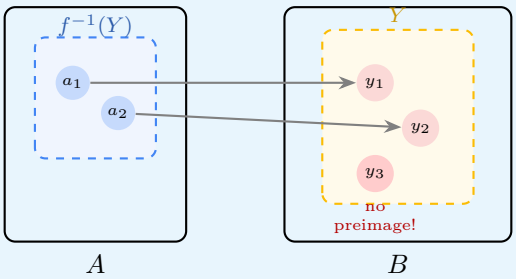
Theory Recap: Image of a Preimage

For $f : A \to B$ and $Y \subseteq B$:

$$f(f^{-1}(Y)) \subseteq Y \qquad \text{always holds.}$$

Equality $f(f^{-1}(Y)) = Y$ holds for all $Y \subseteq B$ if and only if f is **surjective**.

Intuition: $f^{-1}(Y)$ captures all elements in A that map into Y. Applying f again maps them back into Y — but some elements of Y may have *no preimage*, so they are "lost."



Here $f(f^{-1}(Y)) = \{y_1, y_2\} \subsetneq \{y_1, y_2, y_3\} = Y$.

Problem Statement

Prove that for $f : A \to B$ and $Y \subseteq B$:

$$f(f^{-1}(Y)) \subseteq Y.$$

Give an example where equality does not hold.

Solution

Proof of $f(f^{-1}(Y)) \subseteq Y$.

Let $y \in f(f^{-1}(Y))$. By definition of image, there exists $x \in f^{-1}(Y)$ such that $f(x) = y$. By definition of preimage, $x \in f^{-1}(Y)$ means $f(x) \in Y$. Therefore $y = f(x) \in Y$. $\square$

Example where equality fails.

Let $A = \{1\}$, $B = \{a, b\}$, and define $f : A \to B$ by $f(1) = a$. Take $Y = \{a, b\} = B$.

Then:

$$f^{-1}(Y) = f^{-1}(\{a, b\}) = \{x \in A \mid f(x) \in \{a, b\}\} = \{1\}.$$

$$f(f^{-1}(Y)) = f(\{1\}) = \{f(1)\} = \{a\}.$$

But $Y = \{a, b\}$. So:

$$f(f^{-1}(Y)) = \{a\} \subsetneq \{a, b\} = Y.$$

The element $b \in Y$ has no preimage under f (since f is not surjective), so it is "lost."

Proof: If $y \in f(f^{-1}(Y))$, then $y = f(x)$ for some $x \in f^{-1}(Y)$, which means $f(x) \in Y$, i.e. $y \in Y$.
Counterexample to equality: $A = \{1\}$, $B = \{a, b\}$, $f(1) = a$, $Y = B$. Then $f(f^{-1}(Y)) = \{a\} \subsetneq \{a, b\} = Y$.
Equality holds for all Y iff f is surjective.

Companion fact. The reverse inclusion $X \subseteq f^{-1}\big(f(X)\big)$ always holds, with equality if and only if f is injective. Together with this problem's result $f\big(f^{-1}(Y)\big) \subseteq Y$, these two inclusions capture how images and preimages interact.